

Xianglei Dong

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EDUCATION

- University of Detroit Mercy, Bachelor of Engineering** Sep. 2025 – May 2026
Robotic & Mechatronic Systems Engineering; Dean's List (Fall 2025, Winter 2026) GPA: 3.70/4.00
- **Relevant Coursework:** Autonomous Mobility Robotics, Sensors and Actuators, Embedded Systems, Controls I, Robotics, Mechatronics Modeling & Simulation, and Computational Intelligence Techniques.
- Beijing University of Chemical Technology, B.Eng. in Mechanical Engineering** Sep. 2022 – Jul. 2025
Coursework: Signals and Systems, Basic Theory of Mechanical Design GPA: 3.60/4.33

EXPERIENCE

- Hardware and Software Engineer** Jun. 2024 - Sep. 2024
STATE GRID INTELLIGENCE TECHNOLOGY CO.,LTD Jinan, China
- Independently responsible for the design and implementation of the hardware and software architecture for robotic projects, including system integration, sensor calibration, and actuator control.
 - Successfully integrated multiple modules within the project, including real-time operating systems (RTOS), communication protocols (based on ROS and RTThread), and embedded systems, enabling efficient task execution and seamless interaction between modules.
 - Demonstrated excellent project management and problem-solving skills by addressing complex hardware-software integration challenges during testing and deployment.

PROJECTS

*: *Equal Contribution*

- PPO Variants for Humanoid Motion Tracking** Early 2026
Xianglei Dong
- Trained and evaluated five Proximal Policy Optimization (PPO) variants for a 23-joint Unitree G1 humanoid model utilizing **NVIDIA Isaac Lab** and **Isaac Sim** to robustly track complex human reference motions.
 - Leveraged the **Isaac Lab** framework within the Omniverse ecosystem to build highly parallelized simulation environments, bridging the gap between kinematic animations and closed-loop Sim2Real strategies.
 - Developed **Optimized Delta A PPO v2** by decoupling the residual and baseline networks and integrating a pre-trained baseline, which mitigated gradient conflicts during training.
 - Achieved strong training stability (coefficient of variation: 0.068) and a mean reward of 5.03, optimizing the simulation pipeline to complete 6,000 iterations in 2.5 hours on an RTX 4090 GPU.
- Reproducing and Training the ACT Neural Network from the LeRobot Project** Nov. 2024 - Dec. 2024
Jiawei Xia, Xianglei Dong**
- The robotic arm was 3D printed and physically assembled to achieve a fully functional prototype.
 - Trained a custom dataset to enable the robotic arm to autonomously organize a desktop environment.
- Developed an intelligent home service robot based on ROS** Feb. 2024 – May. 2024
Jiawei Xia, Xianglei Dong, Wengcong Zhang, Boyi Zhang, Yutong Sun
- Developed an intelligent home service robot using Kinova Gen2 and Kinect Azure camera.
 - The robot's functions include guest reception and guidance, voice interaction, object recognition, and room cleaning.
 - Integrated Azure AI Speech SDK for speech recognition and output, implementing simple human-robot voice interaction through a finite state machine.
 - Incorporated YOLOv5 and Kinect DK for object recognition, used Cartographer for 2D indoor mapping, employed IRIS LaMa for robot localization, and implemented local path planning using DWA and global path planning using A*.
 - In 2025, wrote an update to yolov8 3d detection and deployed together with graspnet ROS nodes.

ROS 2-Based Autonomous Ground Vehicle for IGVC (Senior Design)
Xianglei Dong, *Kaisheng Zhang, Boyi Zhang, Boni Zhang*

Aug. 2025 - May. 2026

- Developed the high-level autonomy stack for an IGVC-inspired Pioneer 3DX ground vehicle using **ROS 2 Humble**, **SLAM Toolbox**, **Nav2**, LiDAR, and Kinect RGB-D sensing.
- Configured real-time mapping, tuned the DWB local controller for Pioneer 3DX dynamics, implemented frontier exploration, and maintained a custom ROS 2 Humble driver for the robot platform.
- Integrated lane detections into the Nav2 costmap as virtual obstacles and refactored the safety interface to handle asynchronous motion commands more reliably during full-stack testing.
- Validated system performance with **-0.077 s** TF delay, zero SLAM frame drops after queue optimization, **12.4 cm** mean point-navigation error, and a stable **10.5 Hz** command loop.

P2OS ROS2 Humble Porting for Pioneer 3DX
Xianglei Dong

Mar. 2026 – Present

- Successfully migrated the legacy **p2os** driver from ROS Noetic to **ROS2 Humble**, enabling modern middleware support for the classic Pioneer 3DX research robot.
- Refactored the build system from Catkin to **Colcon** and updated the C++ source code to comply with ROS2 client library (reclcpp) standards, including node parameters and message pub/sub logic.
- Validated the migrated package through real-world hardware testing, ensuring reliable teleoperation and sensor data integration (Odometry/Battery) via the modern ROS2 ecosystem.
- Open-sourced the project on GitHub, providing the robotics community with a functional bridge for legacy hardware in the ROS2 era.

RViz Plugin for Multi-Point Navigation
Xianglei Dong

Sep. 2024 - Dec. 2024

- Developed a plugin for RViz within the ROS system to enhance the robot's navigation capabilities.
- Enabled multi-point navigation for robots, significantly reducing the need for manual input of coordinate points in RViz.

AWARDS

2025 RoboCup@Home China National
First Prize

Huairou, Beijing
May. 2025

ICAN International Contest of Innovation
Second Prize

Beijing, Beijing
Nov. 2024

2024 RoboCup@Home China National
Second Prize

Quanzhou, Fujian
May. 2024